CI/CD Report — Learn-Jenkins

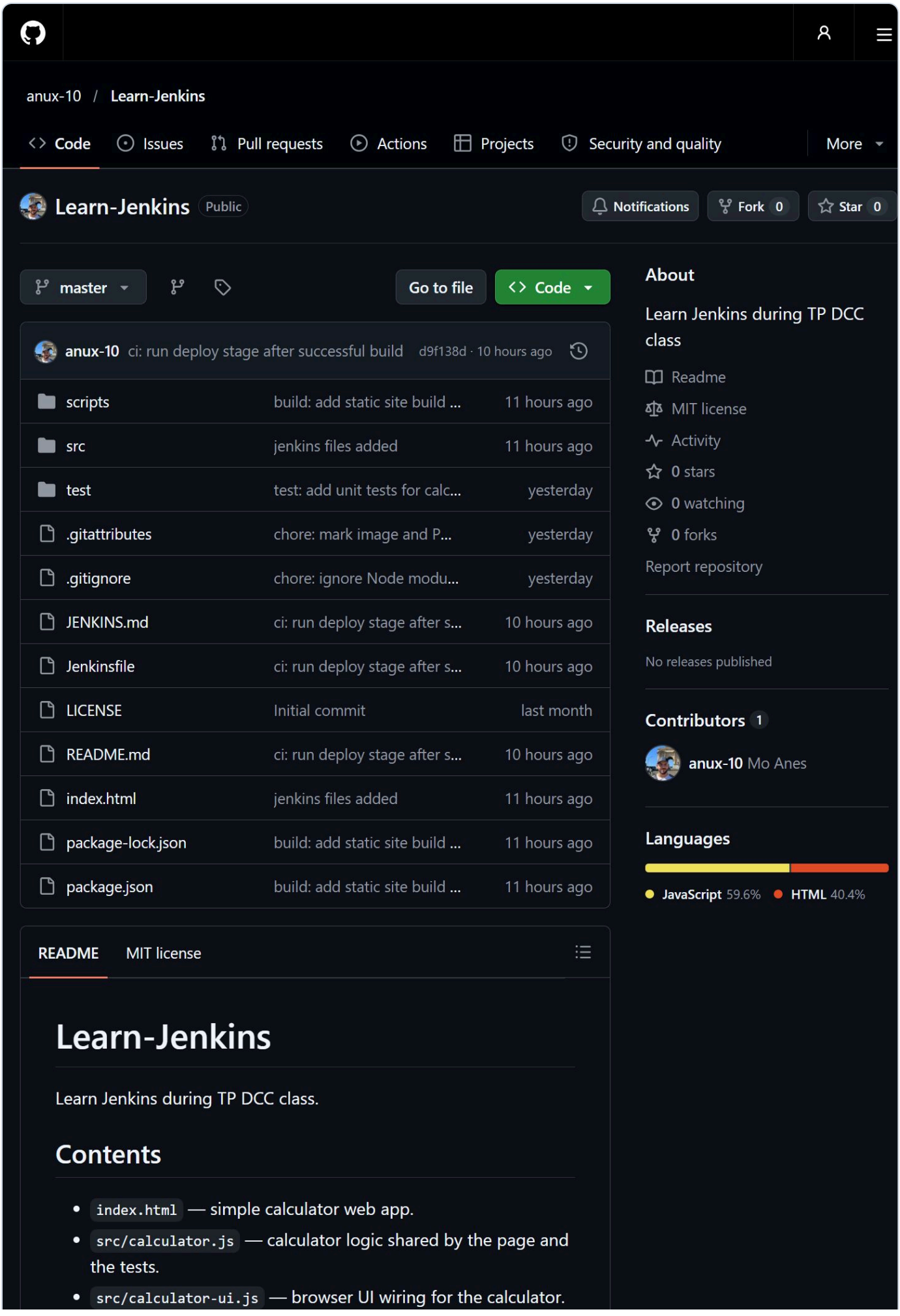
Repository: https://github.com/anux-10/Learn-Jenkins **Jenkins job:** http://localhost:8080/job/Learn-Jenkins/ **Deployed site:** http://localhost:8081/

This report shows the full CI/CD pipeline implemented in Jenkins for the Learn-Jenkins project: repository, Jenkins pipeline, automated tests, and the deployed application.

---

1. Repository

The project lives on GitHub at github.com/anux-10/Learn-Jenkins. Jenkins checks out the `master` branch and reads the `Jenkinsfile` from the repository root.


anux-10 / Learn-Jenkins
Code
Issues
Pull requests
Actions
Projects
Security and quality
More
Learn-Jenkins
Public
Notifications
Fork 0
Star 0
master
Go to file
Code
anux-10 ci: run deploy stage after successful build d9f138d · 10 hours ago
scripts build: add static site build ... 11 hours ago
src jenkins files added 11 hours ago
test test: add unit tests for calc... yesterday
.gitattributes chore: mark image and P... yesterday
.gitignore chore: ignore Node modu... yesterday
JENKINS.md ci: run deploy stage after s... 10 hours ago
Jenkinsfile ci: run deploy stage after s... 10 hours ago
LICENSE Initial commit last month
README.md ci: run deploy stage after s... 10 hours ago
index.html jenkins files added 11 hours ago
package-lock.json build: add static site build ... 11 hours ago
package.json build: add static site build ... 11 hours ago
About
Learn Jenkins during TP DCC class
Readme
MIT license
Activity
0 stars
0 watching
0 forks
Report repository
Releases
No releases published
Contributors 1
anux-10 Mo Anes
Languages
JavaScript 59.6%
HTML 40.4%
README
MIT license
Learn-Jenkins
Learn Jenkins during TP DCC class.
Contents
index.html — simple calculator web app.
src/calculator.js — calculator logic shared by the page and the tests.
src/calculator-ui.js — browser UI wiring for the calculator.

- `test/calculator.test.js` — unit tests using Node's built-in test runner.
- `scripts/test-ci.js` — runs the tests and writes `reports/junit.xml`.
- `scripts/build.js` — builds the static site into `dist/`.
- `package-lock.json` — locked dependency graph used by `npm ci`.
- `Jenkinsfile` — declarative Jenkins pipeline.
- `JENKINS.md` — steps for setting up the Jenkins job.

Run the tests

```
npm ci
npm run build
npm test
```

Or run exactly what Jenkins runs:

```
npm run test:ci
```

Jenkins

The pipeline scans for committed secrets, installs from the lockfile, builds and archives `dist/`, audits dependencies, runs the tests, and publishes the JUnit report. It also deploys the built site to Nginx at http://localhost:8081. See `JENKINS.md` for the job setup steps, deployment details, and security measures.

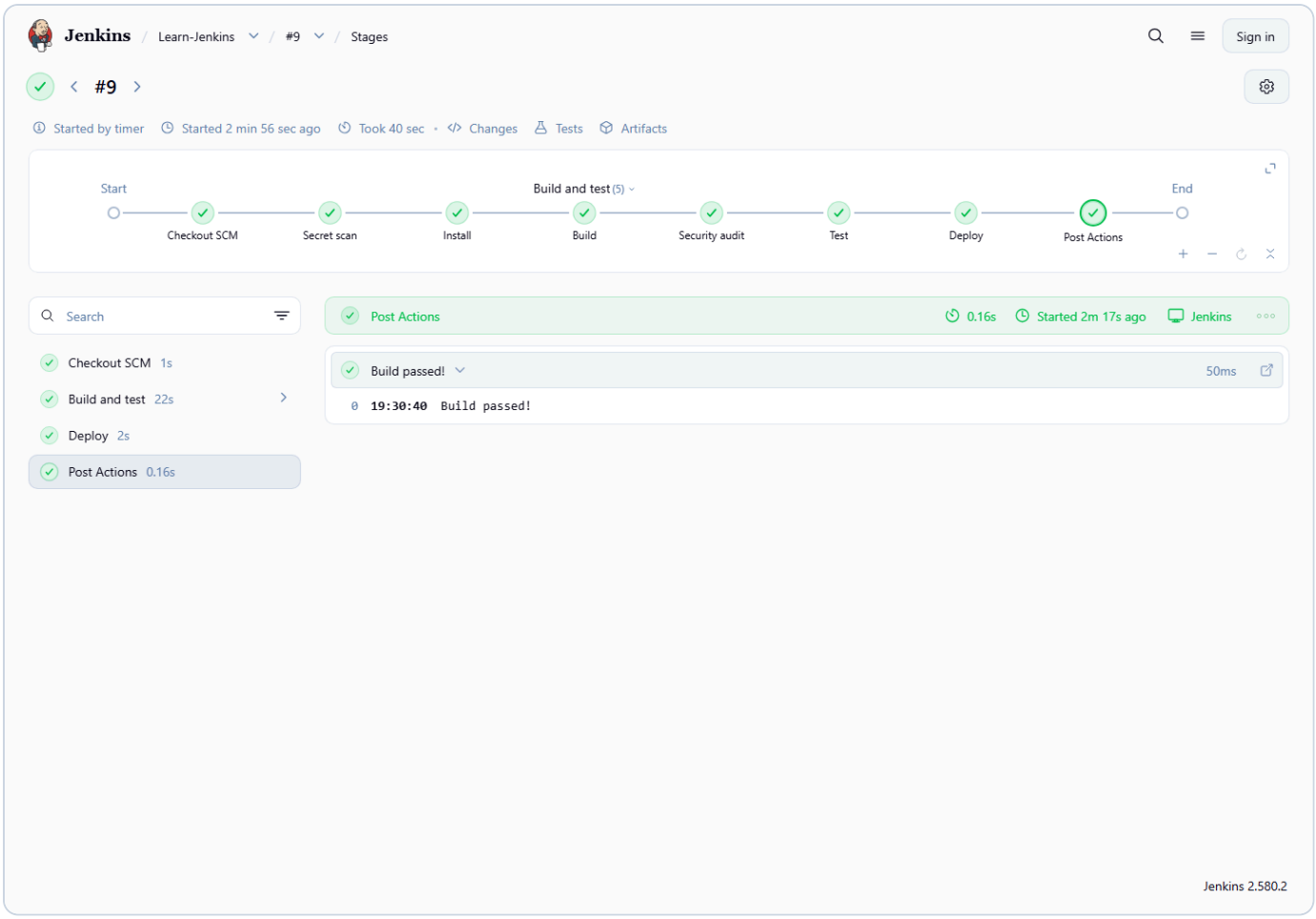
Terms Privacy Security Status Community Docs Contact Manage cookies Do not share my personal information

© 2026 GitHub, Inc.

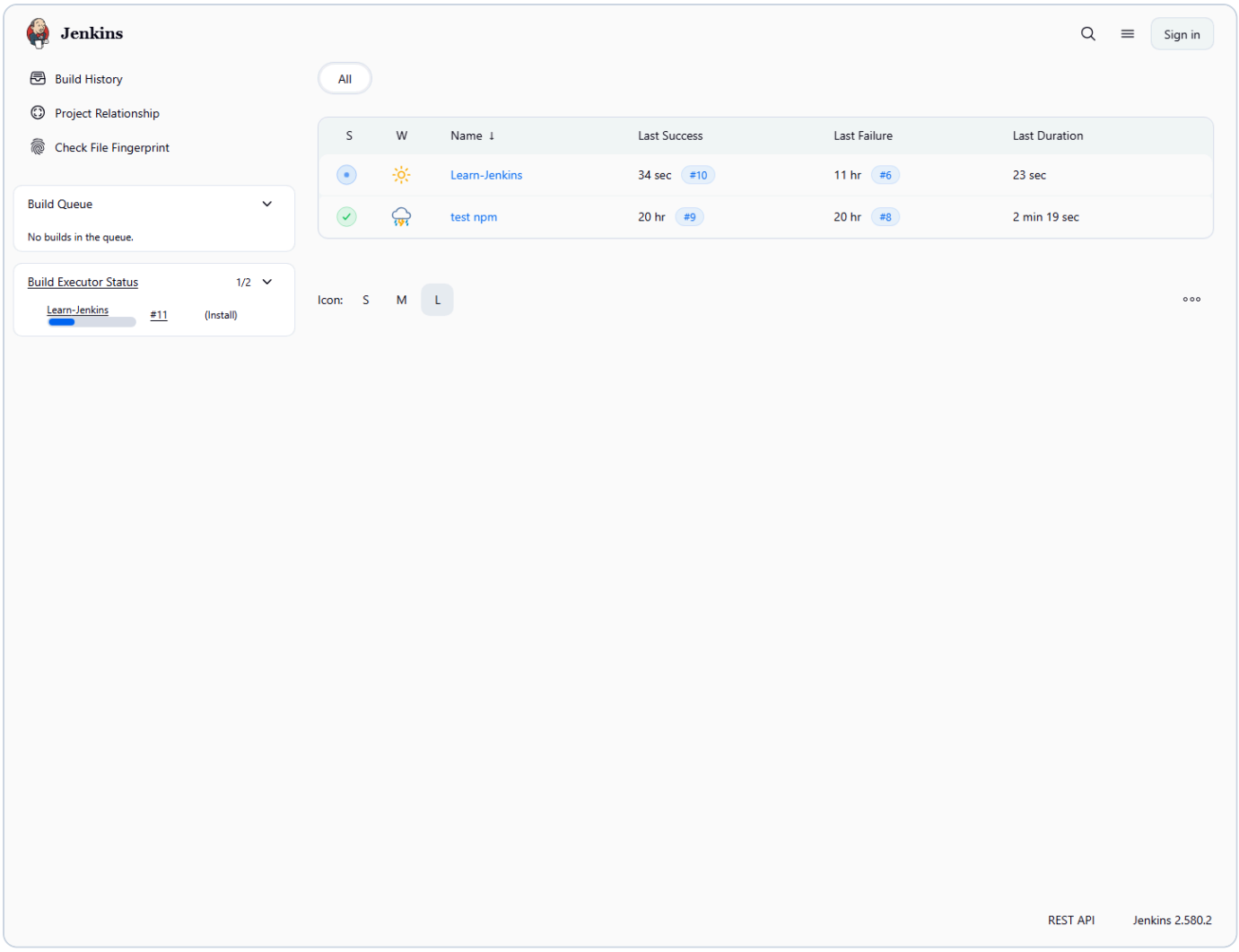
2. Jenkins CI/CD pipeline

The declarative `Jenkinsfile` defines six stages. The latest run, **build #9**, finished in 40 seconds with every stage green, including **Deploy**.

#	Stage	What it does
1	Secret scan	Fails the build if obvious secrets are committed
2	Install	`npm ci` from the lockfile inside a pinned Node 24 container
3	Build	`npm run build` produces `dist/` and archives it as an artifact
4	Security audit	`npm audit --audit-level=high`
5	Test	Verifies `dist/index.html` and runs `npm run test:ci`
6	Deploy	Publishes `dist/` to Nginx at http://localhost:8081

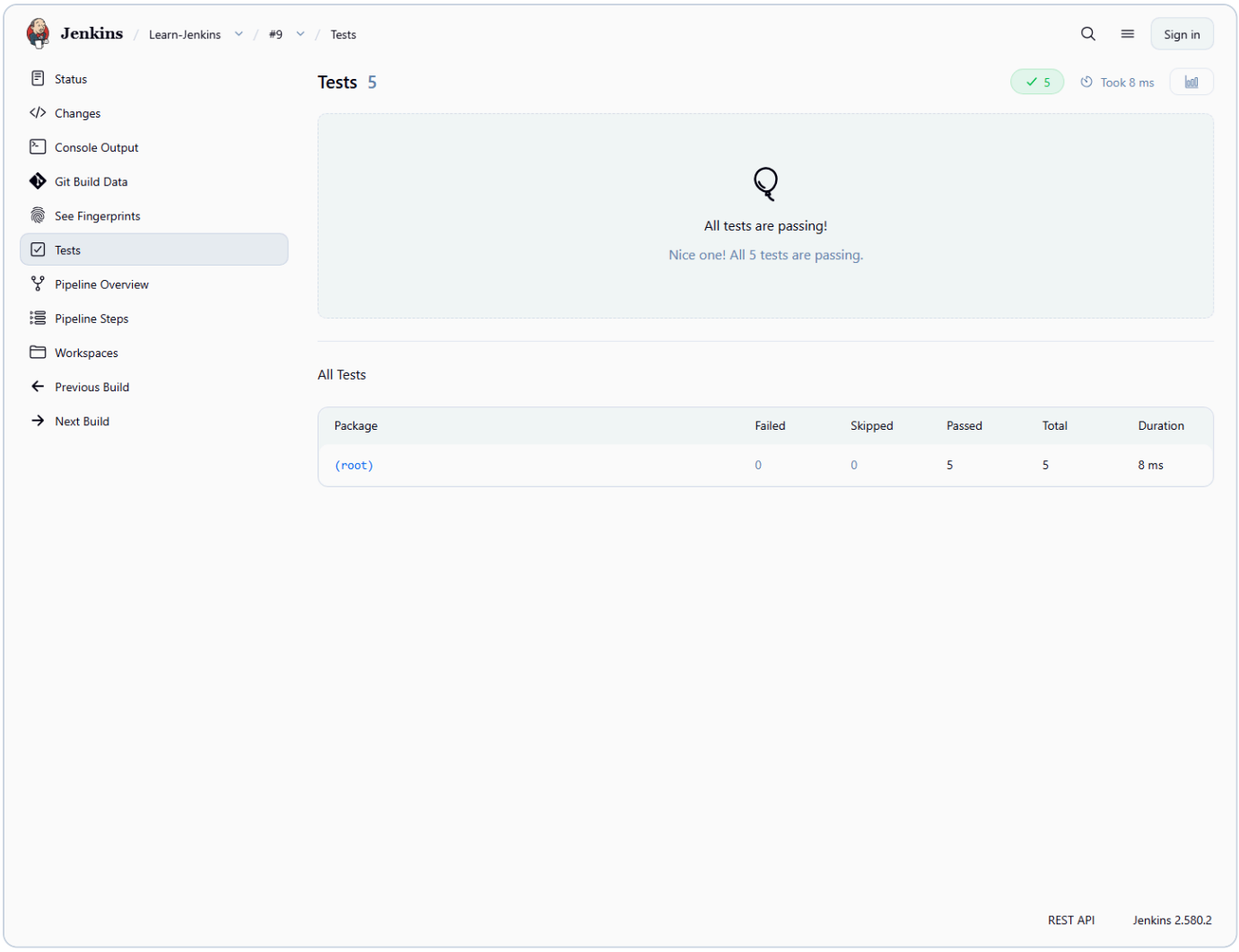


Jenkins dashboard with the job list:



3. Tests

The project uses Node's built-in test runner. Jenkins runs `npm run test:ci`, which executes the suite and writes a JUnit report to `reports/junit.xml`. The report is published on the build page.



Local run of the same command:

```
npm run test:ci
```

```
✓ adds two numbers
✓ subtracts two numbers
✓ multiplies two numbers
✓ divides two numbers
✓ throws when dividing by zero
ℹ tests 5
ℹ pass 5
ℹ fail 0
```

---

4. Deployment

The `Deploy` stage recreates the `learn-jenkins-site` container and copies the built `dist/` folder into it, then checks that the page contains "Simple Calculator".



Item	Value
URL	http://localhost:8081
Container	`learn-jenkins-site`
Image	`nginx:alpine@sha256:713fc2eb…` (pinned by digest)
Restart policy	`unless-stopped`

---

5. Commands used

Trigger a build:

```
curl -X POST http://localhost:8080/job/Learn-Jenkins/build
```

Run the pipeline steps locally:

```
npm ci
npm run build
npm run test:ci
```

Deploy manually (what the Deploy stage runs):

```
docker rm -f learn-jenkins-site
docker run -d --name learn-jenkins-site --restart unless-stopped \
  --security-opt=no-new-privileges -p 8081:80 \
  nginx:alpine@sha256:713fc2ebfd92e8217db451477182ee91944b123d2b1ee9fd5a8c6e91ca6c3a11
docker cp dist/. learn-jenkins-site:/usr/share/nginx/html/
docker exec learn-jenkins-site wget -qO- http://localhost/ | grep -q 'Simple Calculator'
```

Verify the deployment:

```
curl -I http://localhost:8081/
docker ps --filter name=learn-jenkins-site
```

---

6. Security measures

- Pinned Node 24 and Nginx images by digest.
- Containers run with `--cap-drop=ALL` and `--security-opt=no-new-privileges`.
- Build timeout of 15 minutes and `disableConcurrentBuilds()`.
- Secret scan stage blocks obvious credentials.
- `npm audit --audit-level=high` fails on high-severity vulnerabilities.